Climate Dynamics (1994) 10:1–19

Climate Dynamics

© Springer-Verlag 1994

Monte Carlo climate change forecasts with a global coupled ocean-atmosphere model

U. Cubasch[1], B. D. Santer[2], A. Hellbach[1], G. Hegerl[3], H. Höck[3], E. Maier-Reimer[3], U. Mikolajewicz[3], A. Stössel[3], R. Voss[4]

[1] Deutsches Klimarechenzentrum, Bundesstr. 55, D-20146 Hamburg, Germany
[2] Program for Climate Model Diagnosis and Intercomparison, Lawrence Livermore National Laboratory, Livermore, Ca. 94550, USA
[3] Max-Planck-Institut für Meteorologie, Bundesstr. 55, D-20146 Hamburg, Germany
[4] Meteorologisches Institut der Universität Hamburg, Bundesstr. 55, D-20146 Hamburg, Germany

Received: 15 December 1992/Accepted: 2 September 1993

Abstract. Four time-dependent greenhouse warming experiments were performed with the same global coupled atmosphere-ocean model, but with each simulation using initial conditions from different "snapshots" of the control run climate. The radiative forcing – the increase in equivalent CO_2 concentrations from 1985–2035 specified in the Intergovernmental Panel on Climate Change (IPCC) scenario A – was identical in all four 50-year integrations. This approach to climate change experiments is called the Monte Carlo technique and is analogous to a similar experimental set-up used in the field of extended range weather forecasting. Despite the limitation of a very small sample size, this approach enables the estimation of both a mean response and the "between-experiment" variability, information which is not available from a single integration. The use of multiple realizations provides insights into the stability of the response, both spatially, seasonally and in terms of different climate variables. The results indicate that the time evolution of the global mean warming signal is strongly dependent on the initial state of the climate system. While the individual members of the ensemble show considerable variation in the pattern and amplitude of near-surface temperature change after 50 years, the ensemble mean climate change pattern closely resembles that obtained in a 100-year integration performed with the same model. In global mean terms, the climate change signals for near surface temperature, the hydrological cycle and sea level significantly exceed the variability among the members of the ensemble. Due to the high internal variability of the modelled climate system, the estimated detection time of the global mean temperature change signal is uncertain by at least one decade. While the ensemble mean surface temperature and sea level fields show regionally significant responses to greenhouse-gas forcing, it is not possible to identify a significant response in the precipitation and soil moisture fields, variables which are spatially noisy and characterized by large variability between the individual integrations.

1 Introduction

The transient response of the climate system to increasing anthropogenic emissions of greenhouse gases can be simulated reliably only with a global coupled ocean-atmosphere model (O/AGCM; see Houghton et al. 1990). The enormous amount of computer resources required for these experiments, however, has limited the number of institutions which have actually performed such integrations to only four – the National Center for Atmospheric Research (NCAR; Washington and Meehl 1989), the Geophysical Fluid Dynamics Laboratory (GFDL; Stouffer et al. 1989; Manabe et al. 1992), the Max-Planck Institute for Meteorology (MPI; Cubasch et al. 1992) and the United Kingdom Meteorological Office (UKMO; Murphy 1992). At present, even on a state-of-the-art supercomputer, a single 100-year experiment with a fully-coupled O/AGCM requires many months of computer time. Such large demands on computing resources have restricted the total number of long (i.e., >30–100 years) transient greenhouse warming experiments to less than a dozen worldwide.

It is, however, difficult to intercompare these integrations on a common basis, since the models used have different physics, horizontal and vertical resolution, and parameterizations of sub-grid scale processes, and employ different flux correction[1] schemes (Sausen

Correspondence to: U. Cubasch

[1] Even though ocean and atmosphere models do not exhibit a drift (i.e., the mean statistical properties remain time-independent) when run separately, they tend to drift into an unrealistic climate once they have been coupled together. To counter this effect, a flux correction is applied, which allows the models to exchange the information about their respective variability (anomaly coupling), but keeps their mean state close to the observed climatology

et al. 1988; Manabe et al. 1992) in order to minimize drift of the coupled ocean-atmosphere system. In addition, even the time evolution of the greenhouse-gas forcing differs from integration to integration, ranging from a linear increase of 1% yr^{-1} (Washington and Meehl 1989), a compounded 1% increase per year (Stouffer et al. 1989; Murphy 1992) to the non-linear increases specified in scenarios A and D of the IPCC (Cubasch et al. 1992). Model-dependent uncertainties in projecting the climatic response to time-dependent greenhouse-gas forcing are introduced, therefore, not only by fundamental difference in physics, but also by differences in the forcings applied.

It is well-known from the work of Lorenz (1975, 1984, 1991) that the results of simplified climate models consisting of three equations only are highly sensitive to the initial conditions. If this result is transferrable to the world of coupled O/AGCMs, it suggests that sensitivity to initial conditions may constitute one source of model-independent uncertainty in estimating the space-time climate signal in response to greenhouse-gas forcing. We would like to know whether these model-independent uncertainties are large or small in relation to the model's own internally-generated natural variability. We are also interested in determining whether certain climatic fields show lower sensitivity to initial conditions, and thus would be more promising candidates for inclusion in a multivariate "fingerprint" detection strategy.

For example, recent results by Cubasch et al. (1992) and Santer et al. (1993a) from the greenhouse warming experiments performed with the same coupled O/AGCM, but with large differences in the time evolution of the forcing, indicate that land-sea contrast is an important common component of the surface temperature response, and that a similar response pattern emerges in the three integrations. This suggests that the physics associated with the uptake and vertical and horizontal redistribution of heat by the ocean will eventually imprint itself on the surface temperature signal, regardless of the initial conditions of the experiment. If this is the case, the "tube" of all possible surface temperature trajectories in phase space may be increasingly constrained by the physics as the length of the greenhouse warming integration increases.

Hansen et al. (1990) made a first attempt to examine the sensitivity of a climate change signal to initial conditions in experiments performed with an AGCM coupled to a mixed-layer ocean. Such a model cannot be used to investigate the sensitivity to initial conditions of a greenhouse warming signal evolving on time scales of decades to centuries. It is also unsuitable for defining the century time scale natural variability of the coupled atmosphere-ocean system.

Given the long time scales of the deep and intermediate ocean (Mikolajewicz et al. 1990) and the possible existence of normal modes of the ocean circulation with time scales of centuries (Mikolajewicz and Maier-Reimer 1990), it would be necessary to perform very long coupled integrations (>1000 years) in order to define reliably the statistical properties of the century-time scale natural variability of the coupled atmosphere-ocean system. Experiments of such length are prohibitively expensive in computational terms, and thus most control integrations are generally of length 100–200 years only – too short to obtain reliable estimates of century-time scale natural variability. The implication is that the natural variability estimates derived from a single 200-year coupled model integration may themselves be sensitive to the initial conditions, particularly relating to the state of the intermediate and deep ocean. For example, if something similar to the 320-year ocean circulation mode identified by Mikolajewicz and Maier-Reimer (1990) in an experiment involving a stochastically-forced OGCM actually exists in a fully-coupled system, the variability stimulated in a 200-year coupled control run could depend critically on the phase of this mode at the start of the experiment.

The problem of defining a greenhouse signal relative to the natural variability of the coupled ocean-atmosphere system is compounded by the fact that comprehensive four-dimensional data sets of the observed state of the ocean circulation are not available. All coupled model simulations start with an ocean which has been spun up for hundreds to thousands of years in order to represent present-day mean conditions accurately. Given our rudimentary knowledge of present-day conditions in the intermediate and deep ocean, validation of ocean model performance is largely restricted to surface properties only. It is virtually impossible to determine whether the sub-surface initial conditions in the ocean model coincide with the present 'observed' state of the ocean.

A single simulation, therefore, may be unrepresentative of the "true" or underlying space-time greenhouse warming signal, and a single control run of several centuries duration may constitute an inadequate reference state for defining the low-frequency noise characteristics required in order to estimate signal detection times (Santer et al. 1993b). Studies of individual integrations have also shown that the large internal variability of the modelled (and of the natural) climate system can make it difficult to detect a greenhouse-gas climate signal, particularly for spatially-noisy variables such as precipitation rate (Santer et al. 1993a). It may be feasible to enhance signal-to-noisy ratios for such variables by using multiple integrations to reduce both signal and noise uncertainties.

The sensitivity of signal and noise estimates to initial conditions is of course only one source of uncertainty in predicting the response of the climate system to greenhouse-gas forcing. Other uncertainties may be far larger. These include uncertainties related to parameterizations of physical processes, the coupling of atmospheric and oceanic model components, and the possible omission of feedback mechanisms relevant to greenhouse-gas-induced climate change. It is beyond the scope of this study to investigate the sensitivities of signal and noise to basic uncertainties in parameter estimates. While "parameter space" studies can be performed for simpler models (Zebiak and Cane 1987),

they await a future generation of supercomputers and storage technology before they can be performed with global coupled atmosphere-ocean GCMs.

The focus of the present study is on the (model-independent) uncertainties in our predictions of the climate system's response to greenhouse-gas forcing which result solely from imperfect knowledge of the initial conditions of the coupled system. By initial conditions we mean the full, three-dimensional multivariate vector characterizing the state of the atmosphere and ocean at that point in time when greenhouse-gas forcing commences. We consider signal uncertainties only, and do not attempt to investigate the sensitivity of control run variability to initial conditions. We have performed four time-dependent greenhouse warming experiments with the same global, coupled atmosphere-ocean model. Each simulation takes its initial conditions from different "snapshots" of the control run climate. The radiative forcing is identical in all four integrations (as are all other parameters), and is the increase in equivalent CO_2 concentrations from 1985–2035 specified in the Intergovernmental Panel on Climate Change (IPCC) scenario A ("Business as Usual"; Houghton et al. 1990). This approach, known as the "Monte Carlo method" in numerical weather prediction (Brancovic et al. 1990), provides some idea of the statistical properties of the climate system's response to greenhouse-gas forcing.

In section 2, we briefly describe some technical details of the coupled atmosphere-ocean model and the ensemble of four greenhouse warming experiments. Section 3 first examines the sensitivity of the results to the precise definition of the climate change signal, and then considers the regional changes in surface temperature, sea ice, the hydrological cycle, and sea-level. A brief summary and conclusions are given in Section 4.

2 The coupled model and greenhouse experiments

a) The model

The global coupled ocean-atmosphere model used in this experiment has been described in Cubasch et al. (1992). The atmospheric component comprises 19 levels in the vertical and has a horizontal resolution of T21 (on a 5.625° lat./long. Gaussian grid). It has been coupled synchronously to the Large Scale Geostrophic (LSG) ocean model with a horizontal grid spacing similar to the Gaussian grid of the T21 model and 11 layers in the vertical (Maier-Reimer et al. 1993). The coupled model has been used in a number of climate change experiments (Bakan et al. 1991; Cubasch et al. 1992). The control and the Scenario A simulations described in the latter paper form the basis for the experiments introduced here.

b) The Monte Carlo experiments

As stated in the introduction, we are interested in the model-independent uncertainties in a predicted greenhouse-gas signal which result solely from imperfect knowledge of the initial conditions of the coupled system. In the present paper we investigate this sensitivity using the "Monte Carlo" (MC) method, which involves running a suite of climate change experiments with identical forcing but starting from differential initial conditions. A similar method has been applied in extended range numerical weather prediction experiments (Molteni et al. 1988; Brankovic et al. 1990).

Here, we used the 100-year control simulation with the ECHAM-1/LSG model described in Cubasch et al. (1992) in order to obtain initial conditions for the MC experiments. The four greenhouse warming experiments were started at 30 year intervals, with initial fields corresponding to years 0, 30, 60 and 90 of the control experiment (Fig. 1), and with identical radiative forcing taken from the IPCC scenario A experiment (Houghton et al. 1990). All four experiments were run for 50 years, starting at an equivalent atmospheric CO_2 concentration representative of levels in 1985. The integrations are referred to henceforth as MC00, MC30, MC60, and MC90. The first MC experiment (MC00) is identical to the first 50 years of the 100-year scenario A experiment described in Cubasch et al. (1992), here referred to as SZA.

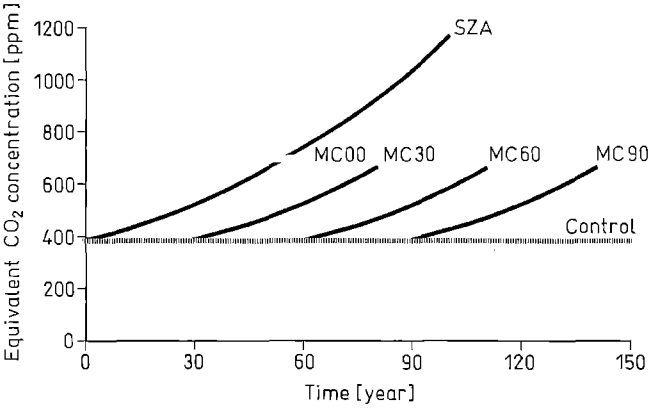


Fig. 1. Schematic diagram of the "Monte Carlo" climate forecasts

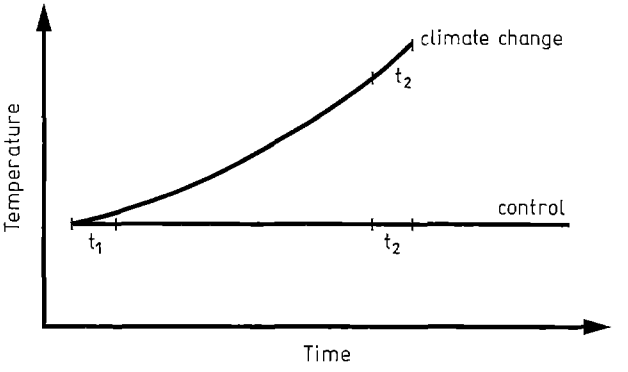


Fig. 2. Schematic diagram of climate change signals according to definitions 1 and 2

3 Results

a) Global mean temperature rise

The temperature at any point in space x and time t of the climate change experiment can be represented by the sum of following terms:

$$T(x,t) = T_0(x) + \Delta T_d(x,t) + \Delta T_f(x,t) + \Delta T_n(x,t)$$

where $T_0(x)$ is the initial temperature, $\Delta T_d(x,t)$ is temperature drift of the simulation (i.e., that component of the experiment's variability which is climate drift attributable to incomplete flux compensation rather than real natural variability of the coupled ocean-atmosphere system), $\Delta T_f(x,t)$ is temperature change caused by the external forcing, and $\Delta T_n(x,t)$ is a realisation of the internally-generated natural variability of the model. Note that it is difficult to discriminate between low-frequency natural variability of the coupled system and residual climate drift if the integration time is short relative to characteristic time scales of deep-ocean variability.

A change in the temperature can therefore be caused by drift, external forcing or internal fluctuations:

$$\Delta T = T_{\text{exp}} - T_{\text{ctl}} = [T_0(x) + \Delta T_d(x,t) + \Delta T_f(x,t) + \Delta T_n(x,t)]_{\text{exp}} - [T_0(x) + \Delta T_d(x,t) + \Delta T_n(x,t)]_{\text{ctl}}$$

Note that $\sigma(x)$, the statistical distribution of natural variability, is assumed to be unchanged by the forcing, while the individual realisations of natural variability may be different.

For calculating the changes of most climate variables, we used definition 1 (c.f. Fig. 2) in accordance with Cubasch et al. (1992). Under this definition, the climate change signal in each Monte Carlo experiment is the difference relative to the smoothed initial state of the control run (at the time the experiment is initialized; i.e., years 1–10, 30–39, 60–69, and 90–99 for MC00, MC30, MC60, and MC90, respectively). The ensemble mean signal is simply the mean of the signals in the four individual experiments. Definition 1 assumes that the variability in the control and response experiments is uncorrelated (i.e., that no drift exists and $\Delta T_d = 0$). We also applied an alternative definition (definition 2), in which the signal in a response experiment is expressed relative to the instantaneous state of the control run. This definition is more appropriate if the control run and response experiment show common residual drift or low-frequency natural variability.

The final choise of either definition is somewhat arbitrary, because the control run and response experiments show features which are consistent with both correlated residual drift and uncorrelated natural variability interpretations (Santer et al. 1993a). In this paper, definition 1 has been used as the preferred definition to facilitate comparison of results with Cubasch et al. (1992). However, for the sea level change and the change in sea ice, definition 2 appeared more appropriate due to the obvious long periodic variability in these quantities (cf. Fig. 15a).

As has been pointed by Cubasch et al. (1992), the annually-averaged global mean near surface atmospheric temperature of the control simulation fluctuates with a maximum anomaly of -0.4 K at the end of the 100-year integration (Fig. 3a). There is a distinct decrease in the global mean temperature of ca. -0.3 K between years 40 and 70. Two of the MC integrations start before this decrease, one starts during the decrease, and the final experiment starts at a time when the global annually-averaged temperature is relatively stable.

All MC simulations display only a small warming at the beginning of the integration (Fig. 3b). This phenomenon is at least partly attributable to the experimental procedure, which neglects the effect of prior oceanic warming from the early days of industrialization up to 1985. This "cold-start" problem has been discussed by Hasselmann et al. (1993) and by Fichefet and Tricot (1992). The latter estimated the temperature error due to the cold start to be as large as 30% of the warming signal after 50 years of integration time if the experiment was started at 1985 conditions instead of pre-industrial conditions. Even if the effect of the cold start error is neglected, the IPCC "best estimate" of the temperature response to scenario A forcing (generated with an upwelling-diffusion energy-balance model; see Wigley and Raper 1990) lies just outside the upper 95% bracket (twice the standard deviation) generated by the four MC experiments (Fig. 3b). The lower 95% bracket includes a period of no appreciable temperature rise for more than 30 years.

In the Hasselmann et al. (1993) study of the "cold start" problem, the warming delay in SZA was estimated by a linear response model involving fitted exponentials. We used a similar model for the individual MC integrations as well as for the ensemble mean (see Hasselmann et al. 1993, for further details of the model). Using this approximation, the cold start response function $R(t)$ in a simulation is represented by the formula:

$$R(t) = \alpha/\mu(1 - e^{-\mu t})$$

where α describes the rate of temperature increase per year and μ is the time constant. The characteristic constants are displayed in Table 1.

Due to the short record length (relative to the time scale of the response), the fit of the exponential curves is not very stable. The coefficients α of the fitted exponential functions vary from 0.05 K/a to 0.2 K/a. The characteristic response time $1/\mu$ varies between 8 and 373 years. The largest temperature rise in the ensemble occurs in MC90. This is reflected in a time constant of 8 years and a rate of temperature increase of 0.2 K/a. Conversely, MC30 has an extremely long characteristic time of 373 years and a lowe rate of temperature increase (0.05 K/a).

The temperature of the mean of the MC ensemble has a rate of increase of 0.07 K/a and a characteristic response time of 30 years. Averaging over the four individual experiments reduces noise, and thus the ensemble mean can be better represented by an exponen-

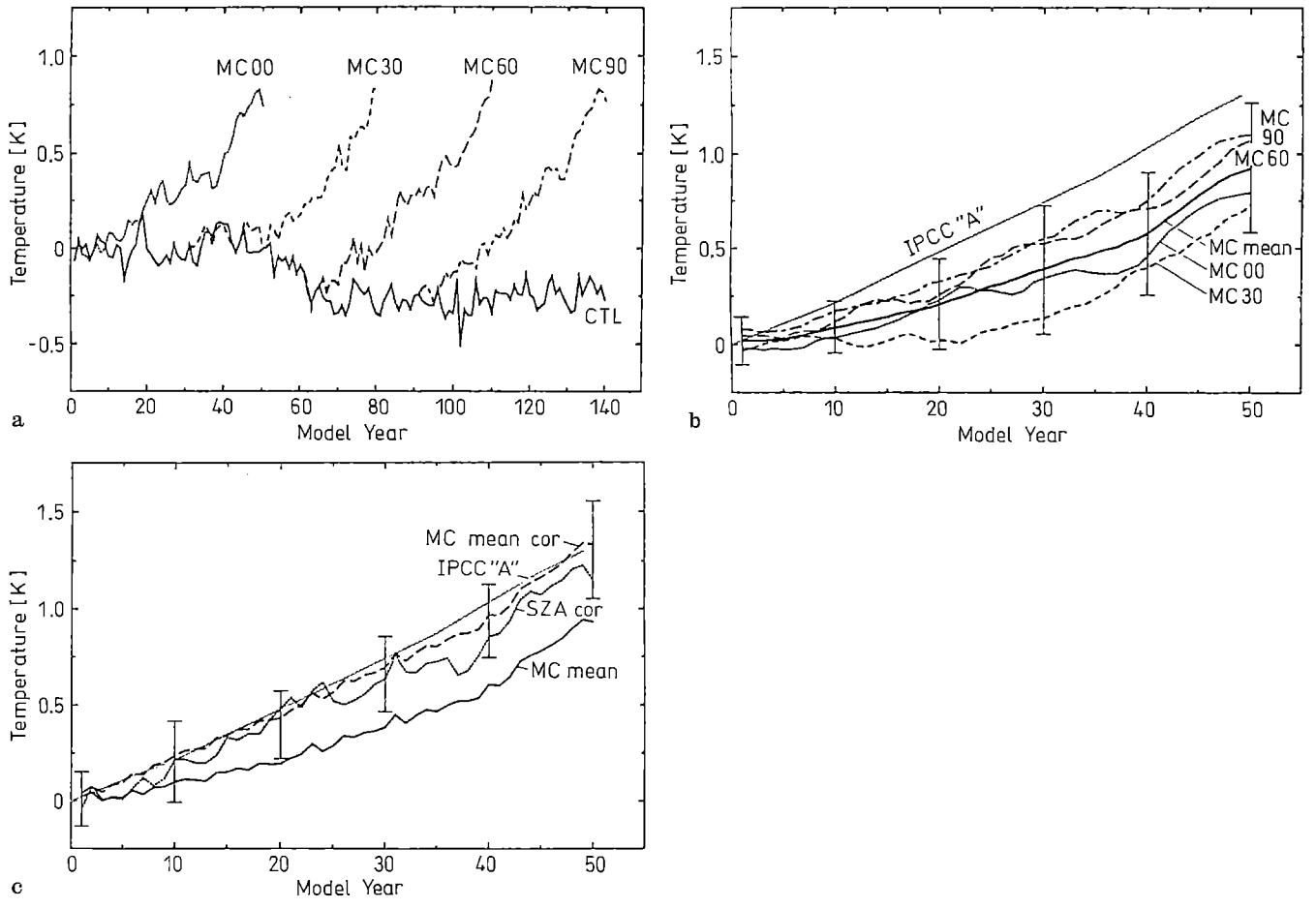


Fig. 3. a Time evolution of the global mean surface temperature change (°C) for the four Monte Carlo simulations and the control experiment. **b** Time evolution of the global mean surface temperature change (°C) for the four Monte Carlo simulations (smoothed), the mean over all four simulations (smoothed), the 95% confidence interval and the IPCC "best estimate" after Houghton et al. 1990. **c** Time evolution of the cold start corrected global mean surface temperature change in the mean over all MC integrations (smoothed), the 95% confidence interval, the cold start corrected curve for the 100-year Cubasch et al. (1992) scenario A experiment and the IPCC "best estimate"

Table 1. The "cold start" characteristics of the "Monte Carlo" climate simulations for the global mean near surface temperature. Detection time estimates for cold start corrected data are given in brackets

Experiment	warming after 50 years [K]	detection time [a]	μ=time-constant [a^{-1}]	α=rate of increase [K/a]	RMS of fit [K]	temperature error at t=infinity [K]
SZA	0.8	29 (15)	1/37	0.06	0.06	0.6
MC00	0.8	29 (15)	1/83	0.05	0.07	1.3
MC30	0.8	37 (25)	1/373	0.05	0.08	7.1
MC60	0.9	23 (11)	1/77	0.06	0.06	1.4
MC90	1.1	20 (11)	1/8	0.20	0.05	0.2
MC mean	0.9	25 (13)	1/30	0.07	0.03	0.5

tial function than the individual members of the ensemble. This is reflected in the low root mean square (RMS) error of the fit, which is only half as large for the ensemble mean as for the individual members.

The exponential fit to the ensemble mean has been used subsequently to estimate the cold start correction, following the approach of Hasselmann et al. (1993). If the individual members of the ensemble are corrected for the cold start, the corrected curves lie both above and below the curve for the IPCC "best estimate". Overall, the spread between the realizations with the highest and lowest temperature changes can be as large as 0.3 K in a single year. Related to an average temperature rise of 0.03 K per year (in the cold start corrected MC mean), this range can be interpreted as an uncertainty in the timing of an temperature change of about

10 years, depending on the initial state of the climate system.

The cold start corrected ensemble mean temperature rice coincides almost exactly with the IPCC's "best estimate" (Fig. 3c). This is at least partly due to the fact that the coupled model and "best estimate" have highly similar climate sensitivities of 2.6 K and 2.5 K, respectively[2] (Houghton et al. 1992).

It is interesting to note that the 100-year scenario A experiment analysed by Hasselmann et al. (1993) is a good proxy for the cold start behavior of the ensemble average. The curve for the cold start corrected MC average lies only slightly higher than the cold start corrected curve of the 100-year scenario A experiment.

A simple measure of the detection time can be defined as the time when the CO_2 warming signal exceeds the mean of the control experiment by two standard deviations (assuming that the drift T_d is negligible and that σ_{ctl} represents the interannual variability of global mean temperature in the control experiment):

$$T_f > T_0 \pm 2\,\sigma_{ctl}$$

The detection time for the non-cold start corrected surface temperatures varies from 11 to 37 years (Table 1). The cold start correction improves the detectability of the warming signal by more than a decade, except in the case of MC90. The detection time for the uncorrected ensemble average is 25 years without cold start correction and 13 years with cold start correction.

To determine the major influences on detection time, we investigated a number of variables whose behavior might explain differences in the rate of warming of the modelled system (e.g., wind stress, ice coverage, and the strength of convective overturning in the oceans. There is some indication that detection time is correlated with ice thickness in the Arctic. However, this correlation appears to be related to the drift in ice thickness, which did not stabilize until year 60 of the control simulation (Cubasch et al. 1992). No clear signal could be identified in any other quantity, probably due to the small sample size.

b) Regional changes

2m temperature. The regional distribution of the temperature change for the last 10 years of the MC ensemble after definition 1 (Fig. 4a) resembles the 2m temperature change pattern of the scenario A experiment described in Cubasch et al. (1992) and Santer et al. (1993a), but with reduced amplitude. It is interesting to note that the regions of cooling in the North Atlantic and North Pacific are a common feature in all integrations and are clearly visible in the mean of the MC ensemble. The smallest standard deviation between the experiments tends to occur in the tropics (Fig. 4c). The maximum between-integration standard deviation is at high latitudes of the Northern and Southern Hemisphere, particularly in the Arctic and Antarctic regions, where it is as large or larger than the temperature change signal. The MC30 experiment shows extensive cooling in the Arctic, while the MC60 and MC90 experiments are characterized by large-scale warming in this area (Fig. 5).

The reasons for these differences in the Arctic temperature signal are at least partially attributable to changes in ice volume during the first part of the control integration (see section on sea-ice). Under definition 1, the Arctic ice volume at the end of the MC00 and MC30 experiments becomes larger than their respective starting values in years 1–10 and 30–39 of the control run (see Fig. 10a), which leads to the Arctic cooling in these experiments. The converse is true for the MC60 and MC90 integrations, where Arctic ice volume decreases under definition 1. If the climate change is calculated according to definition 2, the non-stationarity of the control run's ice volume over the first 60 years is explicitly taken into account, and the Arctic cooling in MC00 and MC30 (under definition 1) is replaced by a slight warming (Fig. 6). This shows that in the presence of substantial control run non-stationarity, the qualitative and quantitative spatio-temporal details of the signal may be sensitive to the precise signal definition.

In the Antarctic the between-experiment differences in the near-surface temperature signal (according to both definitions; see Figs. 5, 6) are mainly related to changes in ice coverage in the Ross and Weddell Seas. The extensive cooling in the Weddell Sea in the final decade of the MC00 (definition 2) and MC30 (definition 1) experiments is related to a sudden decrease in control run ice coverage in this area, with a minimum in year 40 (Fig. 10d).

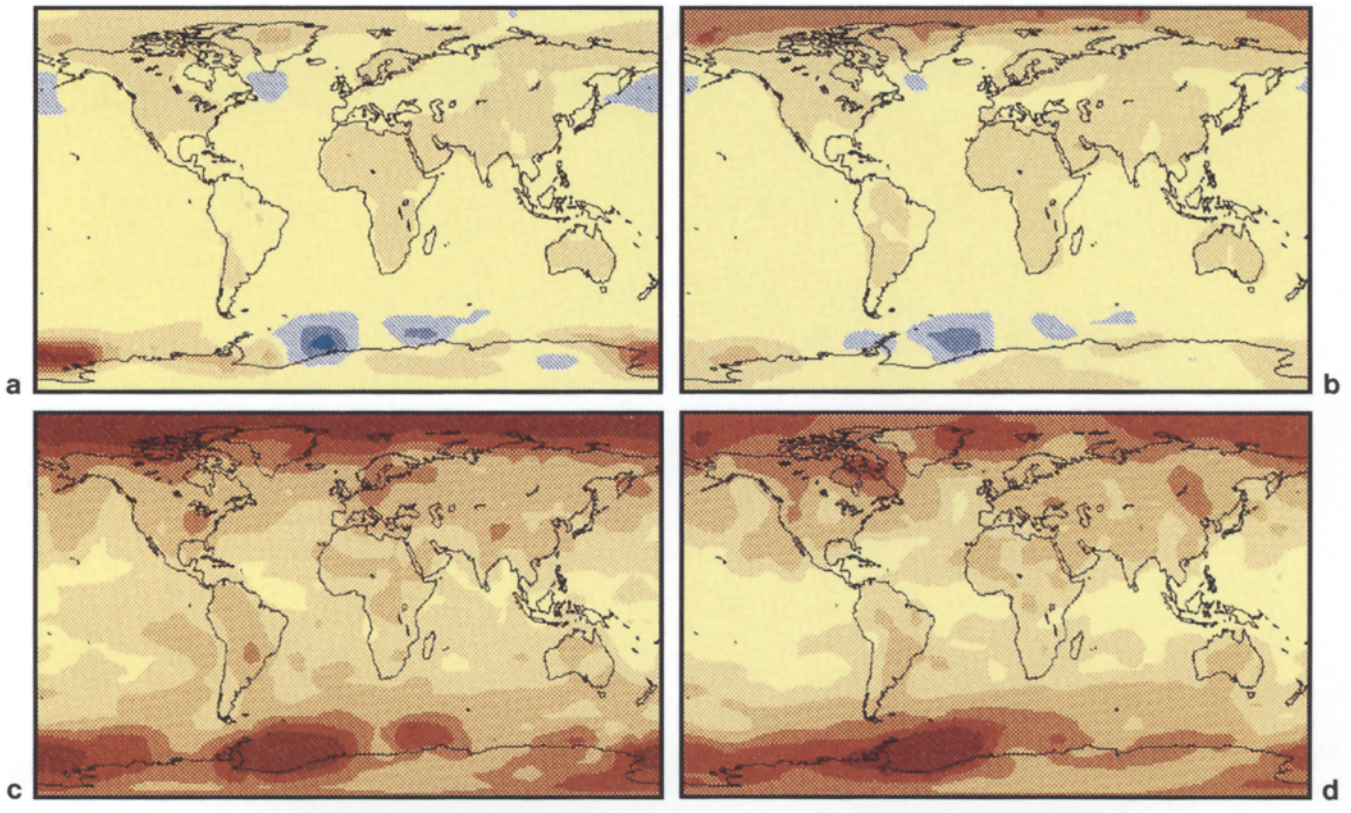
When a *t*-test is applied to individual grid-points, the warming over most of the continents is locally significant at the 95% level, with the exception of a large region stretching from Greece to central Russia (not shown). This suggests that the warming of the continental land masses is a robust feature of the four greenhouse warming experiments. The only regionally significant areas of temperature change over the oceans are located in the central Pacific.

Due to the effects of spatial autocorrelation, individual grid-points are not statistically independent, and it is difficult to judge the global significance of the surface temperature changes on the basis of the cumulative univariate *t*-tests results (Storch 1982; Livezey and Chen 1983; Wigley and Santer 1990). Therefore, we also used a multivariate statistic in order to test the

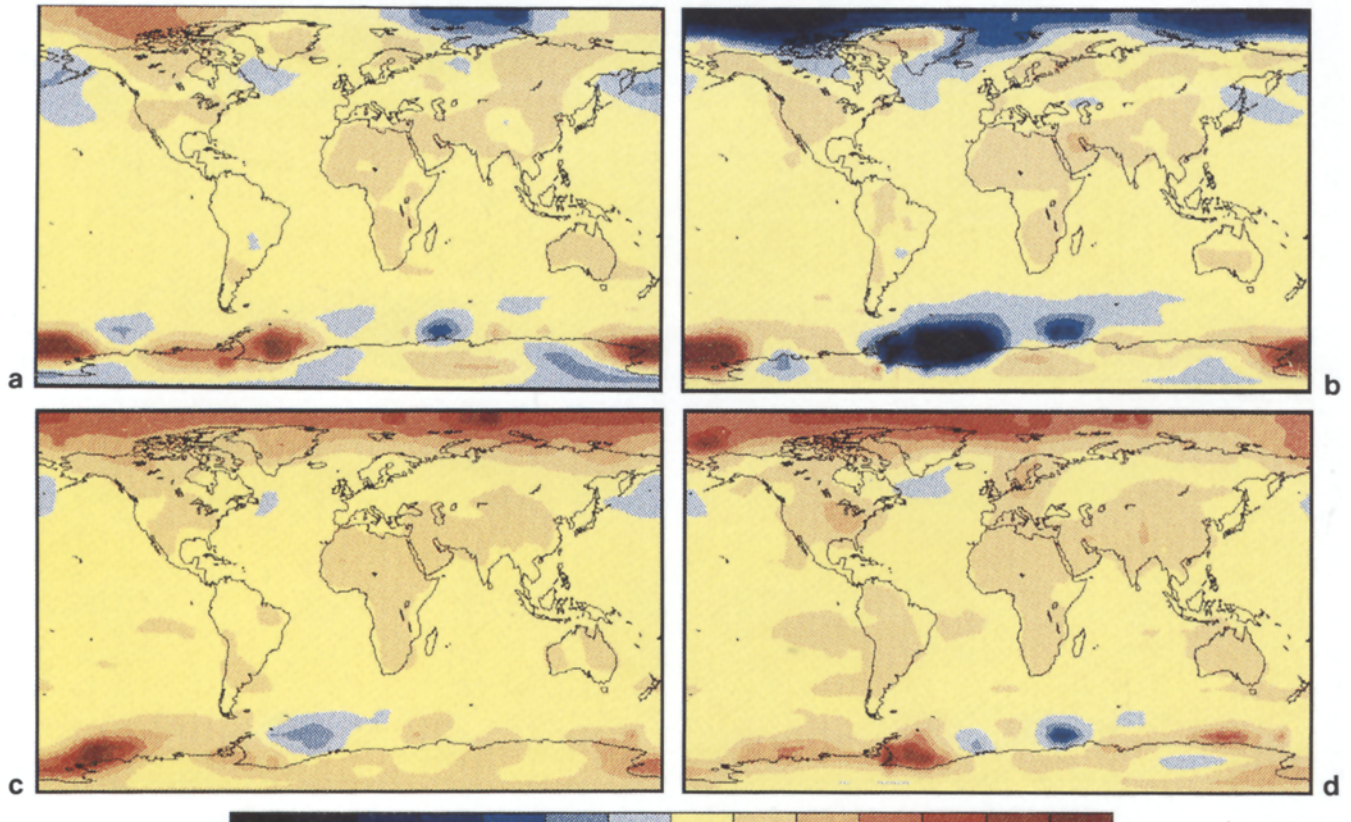
[2] The temperature the model would reach at equilibrium conditions in response to a doubling of CO_2

Fig. 4a–d. Regional distribution of the change in the ensemble averaged annual mean near surface temperature for the final decade of the MC integrations according to **a** definition 1 and **b** definition 2, and the between-experiment standard deviation of the temperature change (**c** definition 1, **d** definition 2)

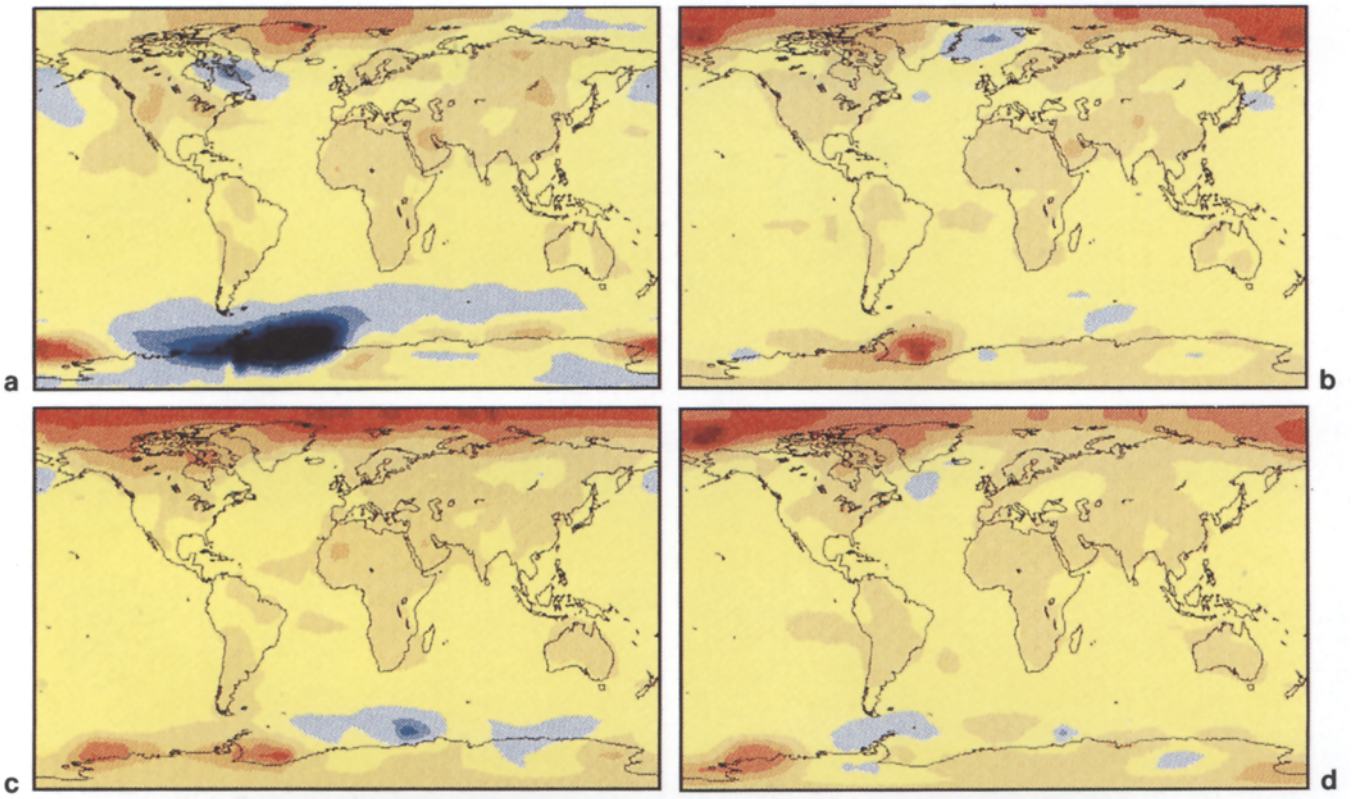
Fig. 5a–d. Regional distribution of the change in annual mean near surface temperature in the final decade of **a** MC00, **b** MC30, **c** MC60 and **d** MC90 integrations. All results are for definition 1


a
b
c
d
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
-5.0 -2.0 -1.0 -0.5 -0.2 -0.1 0 0.1 0.2 0.5 1.0 2.0 5.0

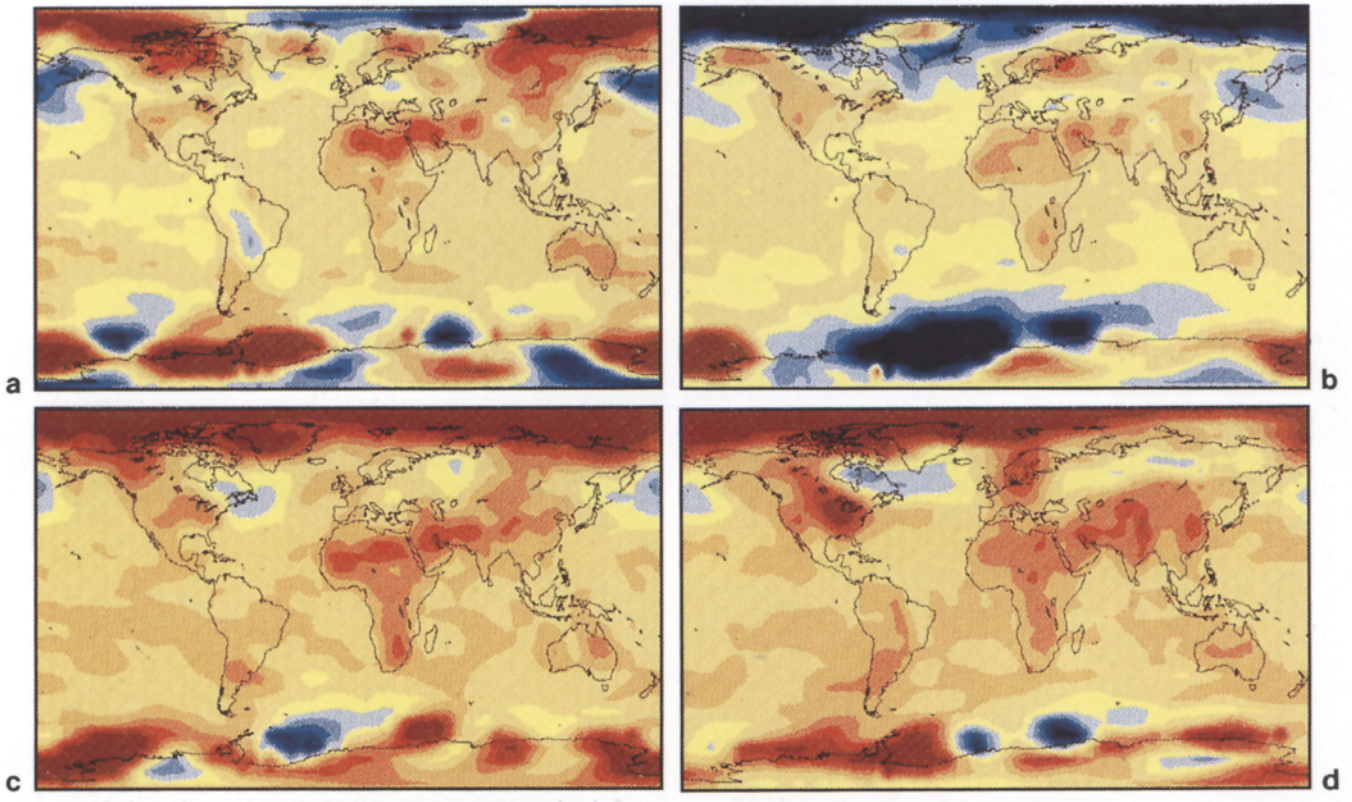
4


a
b
c
d
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6

5



6



7

global null hypothesis of no significant warming. We calculated the vector-product of a spatial "guess" pattern (which defines the properties of the expected global warming signal) with the patterns of the decadally-averaged surface temperature changes in the final decades of the four MC experiments. The "guess" pattern was taken to be the decadally-averaged surface air temperature change in the final decade of the scenario A experiment. Decadally-averaged samples of the control run temperature changes were used to define the statistical properties of the resulting univariate random variable. The vector-products of the four MC signals with the guess pattern exceeds the standard deviation of the sampling distribution (where the sampling distribution is the distribution of the vector-product of guess and control samples) by a factor of approximately eight. We conclude from this that the overall (global) warming of all four MC runs is highly significant.

The regional distribution of the temperature change (definition 1) in the MC experiments also varies considerably with the seasons (not shown). The Antarctic experiences a cooling in the southern hemisphere summer (DJF). This cooling is a robust feature in all MC integrations. In the same season, the Arctic experiences a warming. As in the case of the annual mean, the warming in DJF is influenced by the drift in ice thickness in the control run. The regional cooling near Greenland, which is particularly distinct in the winter season, is simulated in all four MC experiments, and is related to a reduction of deep vertical mixing in the ocean, with a consequent decrease in upward convective heat flux (Mikolajewicz et al. 1990).

In the southern hemisphere winter (JJA) the Antarctic land mass warms up in all integrations. Cooling takes place in the vicinity of the ice edge due to a reduced exchange with water in the deeper ocean layers, while a reduction in ice thickness leads to warming in the Ross Sea. In the MC30 experiment the Ross Sea is cooler than in the control experiment. As in the case of annual mean surface temperature, this is caused by a temporary decrease in the ice thickness of the control experiment. The largest between-experiment variability is at high latitudes of both hemispheres. The seasonal distributions of changes in the surface temperature are generally similar to those described in the IPCC report (Houghton et al. 1992).

EOF analysis of 2m temperature changes. In the following section we describe the results of an EOF analysis of annual mean near surface temperature for each member of the MC integration. For a full description of the analysis procedure, refer to Santer et al. (1993a). EOFs are presented for both definitions of the climate change signal.

Fig. 6a–d. Regional distribution of the change in annual mean near surface temperature in the final decade of **a** MC00, **b** MC30, **c** MC60 and **d** MC90 integrations. All results are for definition 2
Fig. 7a–d. First EOF [$\times 10^2$] of the annual mean near surface temperature change for **a** MC00, **b** MC30, **c** MC60 and **d** MC90 integrations. All results are for definition 1

Independent of the definition, the first EOF (Figs. 7 and 8) shows large between-experiment variability of the signal at high latitudes in both hemispheres. In low and mid-latitudes the patterns in all MC integrations are generally spatially coherent and show a pronounced land-sea contrast component. The temperature changes at high latitudes in both hemispheres are sometimes of opposite sign to changes in the tropics and mid-latitudes, particularly in the case of MC00 (definition 1 and 2) and MC30 (definition 2).

If definition 1 is applied, the total space-time variance explained by the EOF 1 pattern varies between 50% and 67% (see Table 2). This is lower than the variance explained by the dominant pattern in the 100-year SZA integration (84%; see Cubasch et al. 1992), which is simply due to the shorter integration time of the MC simulations. Note that the first EOF of the ensemble mean response (Fig. 9a) explains more of the total space-time variance (78%) than any of the individual MC integrations. When definition 2 is applied (Fig. 8), the first EOF in each of the MC simulations explains less variance than under definition 1 (Table 2). This difference is due to the fact that definition 2 introduces more noise than definition 1 (Santer et al. 1993a). Definition 2 superimposes the interannual variability of the control simulation upon the variability of the climate change experiment, whereas definition 1 considers the interannual variability of the perturbation experiment only.

In order to make a more objective comparison between the EOF 1 patterns of the individual members of the ensemble of MC experiments, we computed area-weighted anomaly pattern correlations with and without subtraction of the spatial mean component (r and r^*, respectively; Santer et al. 1993a). For definition 1, the between-integration pattern correlations (with spatial mean subtracted) do not exceed 0.31, mainly due to pattern differences at high latitudes (Fig. 7 and Table 2). When the spatial mean is incorporated, the large-scale spatial coherence of the EOF 1 patterns substantially improves the pattern correlation (Table 3).

The EOF 1 patterns calculated under definition 2 are generally more highly intercorrelated than under definition 1. This is true for both r and r^* (Tables 2 and 3), and is due to the fact that the EOF 1 patterns computed under definition 2 tend to have a more spatially-coherent structure at high latitudes, particularly in the Northern Hemisphere (cf. Figs. 7 and 8). The lower between-experiment pattern correlations under definition 1 are strongly dependent on the amplitude and spatial coherence of the decadal mean control pattern which is used to calculate the signal anomaly.

The first EOF for MC00 calculated under definition 2 (Fig. 8a) and the first EOF of MC30 calculated under definition 1 (Fig. 7b) both display a very similar pattern in the Southern Ocean. This is caused by a large fluctuation in sea ice coverage in the Weddell Sea in the control experiment between years 30 to 60, resulting in strong warming of the Weddell Sea region in the control experiment. Under definition 1 this anomaly

Table 2. Between-integration pattern correlations for EOF 1 of the individual members of the ensemble of MC experiments and the mean of the ensemble. All correlations are computed with removal of the spatial mean component (r). Results are for definition 1 (upper triangle of matrix) and definition 2 (lower triangle of matrix and italics) of the signal. The explained variance of the EOF 1 pattern under both definitions is also given. The diagonal elements of the matrix (bold) give the correlation between the EOF 1 patterns of the same experiment under the two definitions

Correlation between definition 1 and definition 2	MC00	MC30	MC60	MC90	MC mean	SZA	Explained variance (definition 1)
MC00	**0.02**	0.06	0.20	0.26	0.54	0.76	0.50
MC30	*0.47*	**0.28**	0.31	0.07	0.76	0.47	0.67
MC60	*0.12*	*0.62*	**0.50**	0.26	0.61	0.53	0.56
MC90	*0.30*	*0.73*	*0.80*	**0.69**	0.51	0.44	0.62
MC mean	*0.80*	*0.83*	*0.65*	*0.77*	**0.75**	0.86	0.78
SZA	*0.66*	*0.85*	*0.64*	*0.80*	0.92	**0.56**	0.84
Explained var. (definition 2)	*0.44*	*0.30*	*0.41*	*0.39*	*0.67*	*0.76*	

Table 3. Between-integration pattern correlations for EOF 1 of the individual members of the ensemble of MC experiments and the mean of the ensemble. All correlations are computed without removal of the spatial mean component (r^*). Results are for definition 1 (upper triangle of matrix) and definition 2 (lower triangle of matrix and italics) of the signal. The diagonal elements of the matrix (bold) give the correlation between the EOF 1 patterns of the same experiment under the two definitions

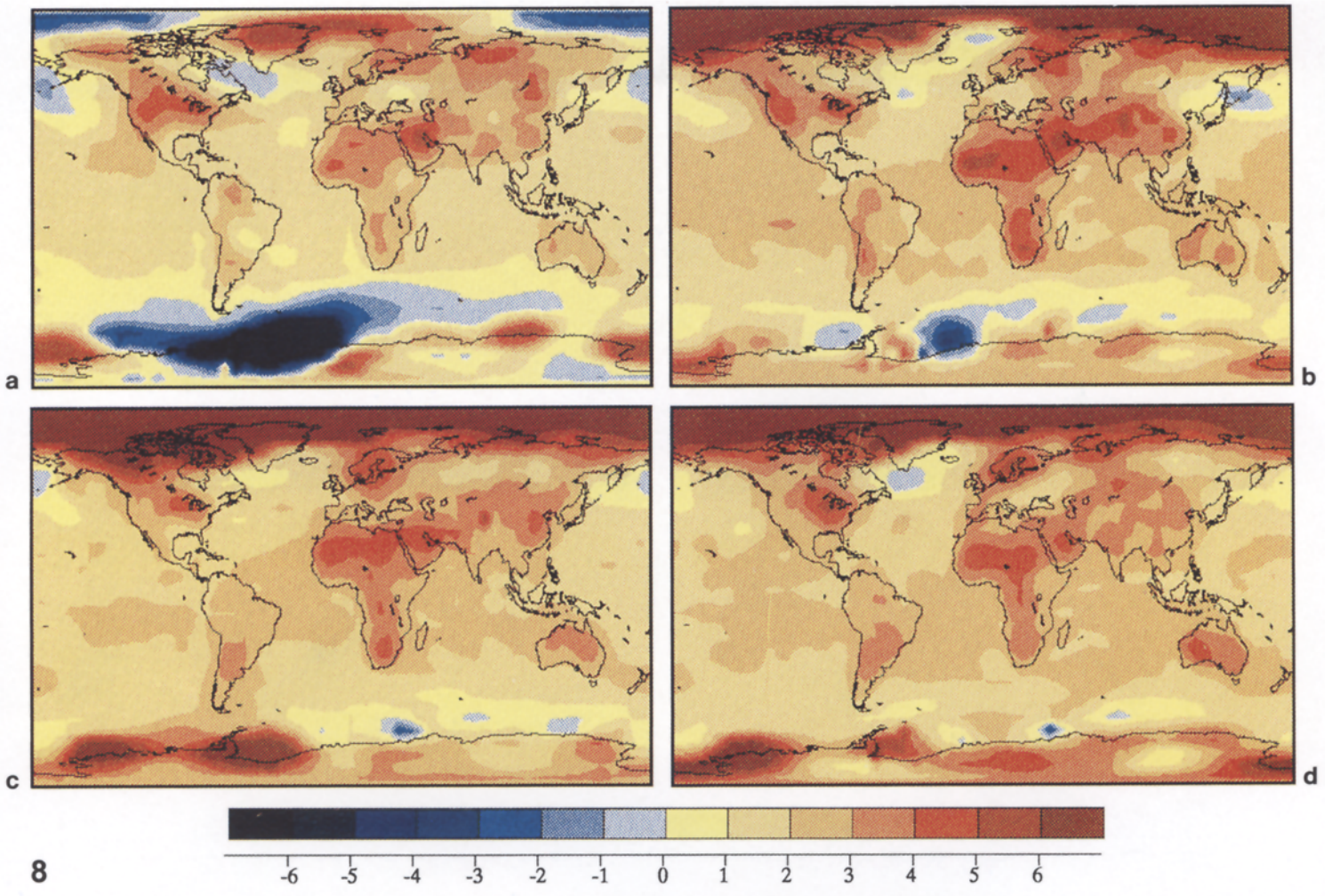
Correlation between definition 1 and definition 2	MC00	MC30	MC60	MC90	MC mean	SZA
MC00	**0.28**	0.16	0.62	0.64	0.76	0.84
MC30	*0.61*	**0.29**	0.31	0.18	0.59	0.39
MC60	*0.45*	*0.91*	**0.88**	0.84	0.87	0.87
MC90	*0.53*	*0.93*	*0.96*	**0.92**	0.83	0.84
MC mean	*0.76*	*0.96*	*0.91*	*0.90*	**0.92**	0.95
SZA	*0.67*	*0.96*	*0.93*	*0.96*	0.98	**0.90**

pattern is projected onto the MC30 experiment (since the decadal average of years 31–40 of the control is subtracted), while under definition 2 it influences the signal in the final decades of the MC00 experiment. This anomalous behavior of the control run during years 30 to 60 explains why the correlation between the EOF 1 patterns generated under both definitions (the diagonal elements of Tables 2 and 3) is small for the MC00 and MC30 experiments, and large thereafter.

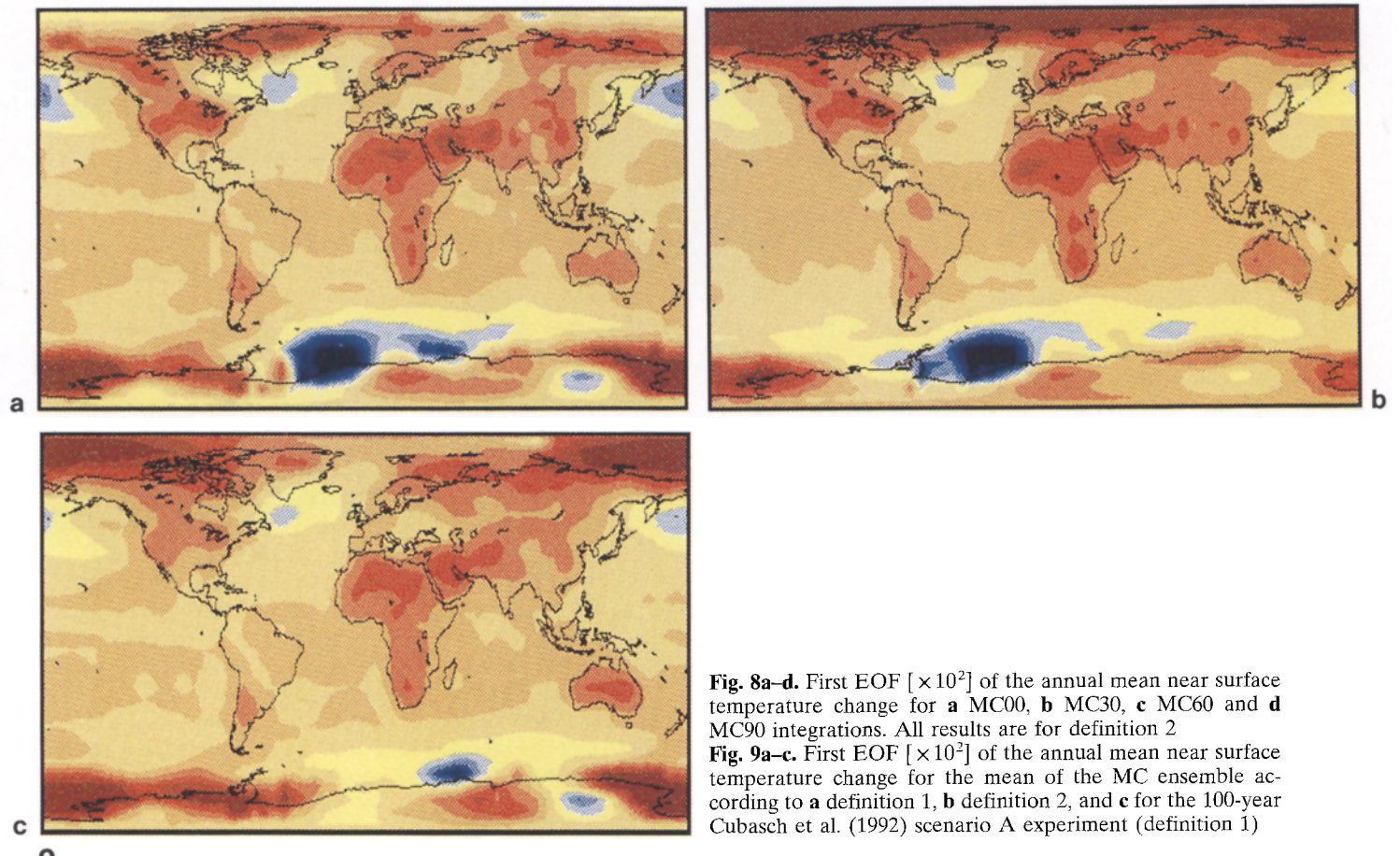
The pattern of the first EOF of the ensemble mean of the MC experiments closely resembles the pattern of the scenario A (SZA) experiment after 100 years ($r=0.86$, $r^*=0.95$; definition 1; Tables 2, 3, Fig. 9a, c). The major difference is in the Weddell Sea, since the previously-discussed control anomaly in years 30–60 influences the signal in the MC30 experiment (and hence the MC mean), but not in the scenario A experiment (when both signals are computed according to definition 1). The ensemble mean of the MC simulations under definition 2 (explained variance: 67%; Fig. 9b) shows a larger signal in the Arctic and reduced variance in the Antarctic relative to definition 1. As for definition 1, averaging the MC ensemble yields a signal which is even more highly correlated with the EOF 1 pattern of SZA ($r=0.92$, $r^*=0.98$; Tables 2, 3). These results, together with the high correlation between EOF 1 of the MC ensemble means under both definitions ($r=0.75$, $r^*=0.92$), show that ensemble averaging is a useful tool to isolate the climate change signal in short and noisy samples.

Sea-ice. The previous sections showed that temperature changes are of largest amplitude in the polar regions mainly due to changes in the sea-ice coverage and thickness. In order to understand better the results of the MC simulations, it is useful to examine whether the coupled model simulates realistic changes in the distribution and thickness of sea ice in the control run.

The most crucial variable for evaluating the skill of sea-ice simulations is the areal sea-ice thickness distribution. Its integrated product is the ice volume. In the control simulation, sea ice requires ca. 100 years in the Northern Hemisphere to equilibrate from an initial annual mean ice volume of about $9 \cdot 10^3$ km^3 to around $40 \cdot 10^3$ km^3 (Fig. 10a). In the Southern Hemisphere, the same simulation yields an ice volume of up to $75 \cdot 10^3$ km^3 after about 70 years starting from an initial annual mean ice volume of $28 \cdot 10^3$ km^3 (Fig. 10c). For comparison, the present-day annual mean sea-ice volume is about $20 \cdot 10^3$ km^3 in the Northern Hemisphere (e.g. Loewe 1987) and approximately 10 to $16 \cdot 10^3$ km^3 in the Southern Hemisphere (Parkinson and Bindschadler 1984; Stössel et al. 1990). Thus, the control simulation yields ice-volume increases of more than



8



9

Fig. 8a–d. First EOF [$\times 10^2$] of the annual mean near surface temperature change for **a** MC00, **b** MC30, **c** MC60 and **d** MC90 integrations. All results are for definition 2

Fig. 9a–c. First EOF [$\times 10^2$] of the annual mean near surface temperature change for the mean of the MC ensemble according to **a** definition 1, **b** definition 2, and **c** for the 100-year Cubasch et al. (1992) scenario A experiment (definition 1)

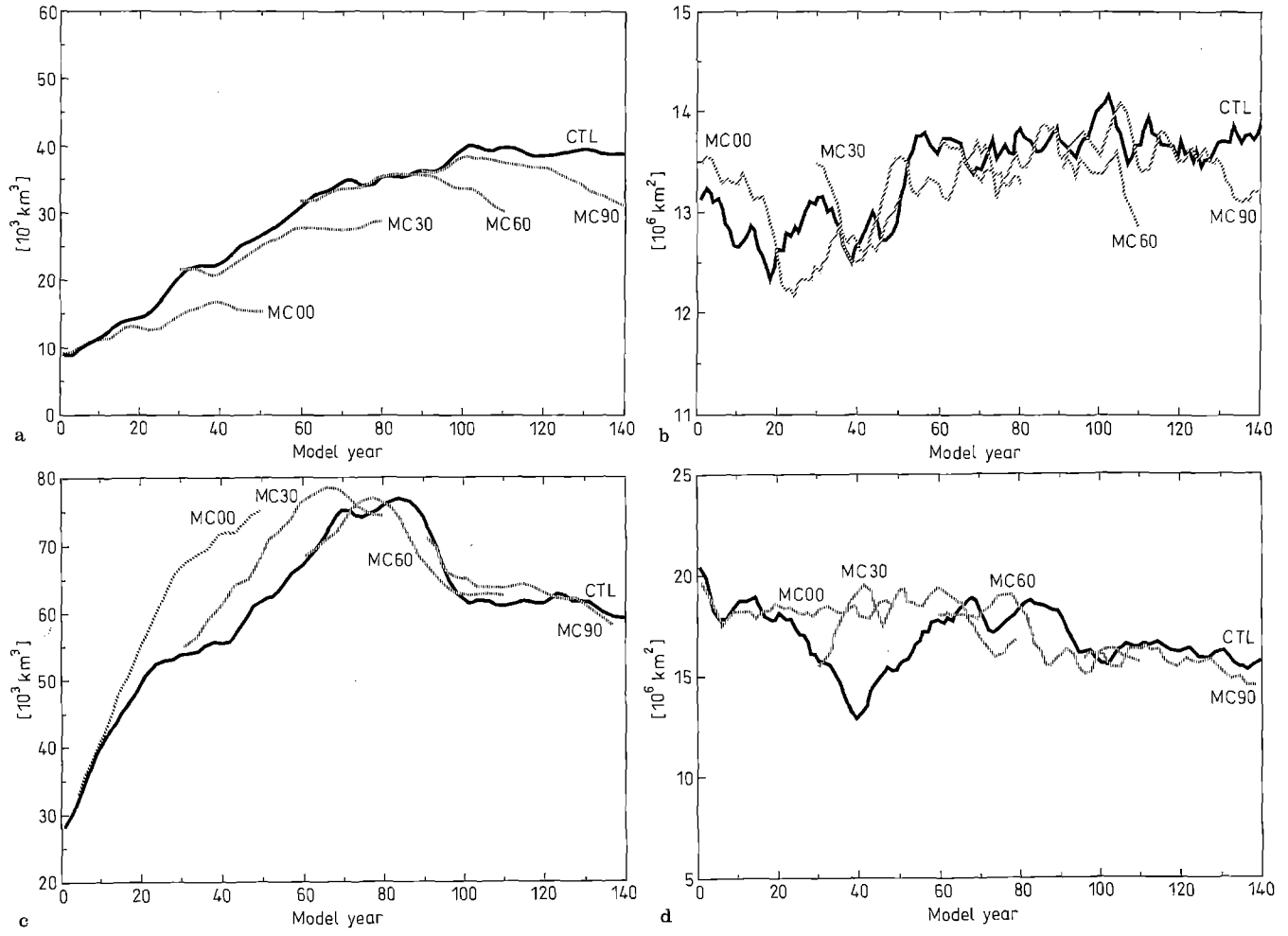


Fig. 10a–d. Time evolution of ice volume in **a** Northern and **c** Southern Hemispheres and the annual maximum ice extent in **b** Northern and **d** Southern Hemispheres

twice and up to four times the initial values (the latter deviating by a factor of two from the present-day observed values). This unrealistic behavior in the simulations makes an evaluation of the results in sea-ice covered regions rather difficult.

In the MC simulations, the ice volume in the Northern Hemisphere generally decreases if definition 2 is applied. This is mainly due to a decrease in ice thickness, as the ice extent does not change significantly during the integrations. In the Southern Hemisphere the ice volume increases in the first experiment (MC00) after application of definition 2, but shows only small changes at the end of the other three MC experiments. The ice extent remains rather stable on long time scales (Fig. 10d). The exception is the previously noted decrease in ice coverage in the control experiment due to a melting of the ice in the Weddell Sea, an event not simulated in any of the MC integrations. This decrease in the ice coverage in the control simulation is more than twice as large as the change in any of the MC simulations.

While the interannual fluctuations in ice extent of the control simulation are within the observed range of variability (Fig. 11; Zwally et al. 1983; Parkinson 1991, 1992), the simulated seasonal variations of ice extent are overestimated in the Arctic Ocean and severely underestimated in the Southern Ocean. The latter error originates from isolated patches where ice tickness is strongly overestimated. As a result, while the simulated ice extent coincides with observations in the Southern Hemisphere winter, the large ice thickness prevents sufficient ice melt in summer. The summer ice extent is therefore reduced by only 23% compared to winter, while in the observations reductions of up to 80% are common. In the MC simulations (not shown), the ice-melt in summer is increased, while the winter ice extent remains the same. The seasonal cycle is therefore enhanced under climate change conditions.

In the Arctic Ocean, on the other hand, the seasonal cycles of ice extent are overestimated in the control integration. The summer ice extent is about one third of the observed value (Fig. 11a). The increase of ice volume during the control integration (Fig. 10a) is primarily caused by local increases in ice thickness, which eventually reaches values which prevent the ice from melting in summer. Since these increases are confined

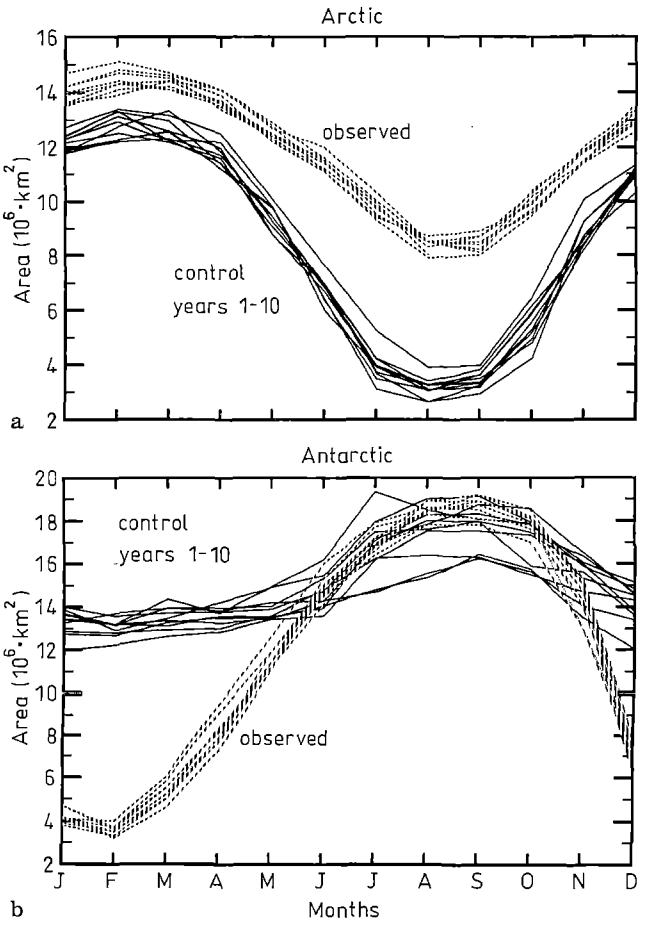


Fig. 11a, b. Annual cycle of **a** Arctic and **b** Antarctic ice extent in the first 10 years of the control simulation and as derived from satellite microwave data by Gloersen and Campbell (1988) representing the years 1978–1987

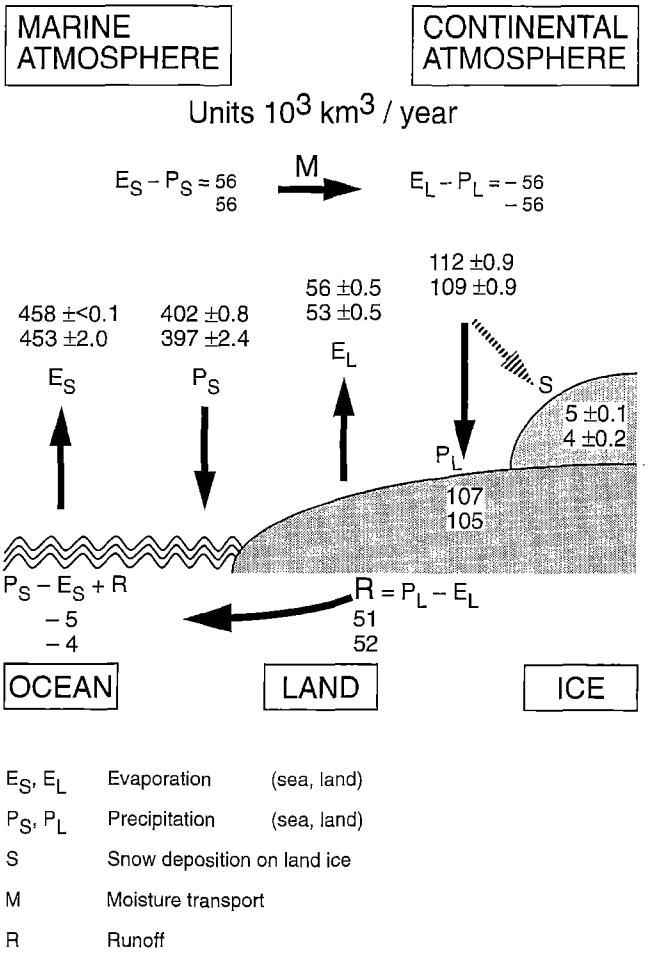


Fig. 12. The hydrological cycle in the control experiment and in the ensemble mean of the MC integrations. Top figures are for the MC ensemble mean, bottom figures for the control experiment

to isolated grid-points, the summer ice extent shows little change during the integration (ca. $5 \cdot 10^6$ km^2 in years 91–100 of the control run). The seasonal cycle of ice extent in the Northern Hemisphere does not change significantly in the Monte Carlo simulations.

The problems with the sea-ice simulation are apparently related to both the application of the flux correction and the crude physical treatment of the sea-ice component. The flux correction has been introduced to minimize the climate drift in a coupled ocean-atmosphere model. Similar correction methods are used by other modelling groups (e.g., Manabe et al. 1992). All are essentially based on a linearization of the errors of the fluxes exchanged between ocean and atmosphere (Sausen et al. 1988) and are therefore only applicable in the case of small changes, where a linearization is valid.

At the ice edge, however, the differences in the fluxes are large in the transition zone between ice-free and ice-covered points and therefore violate the inherent assumptions of the flux correction. Specifically, large heat fluxes may occur due to flux corrections in grid cells where the uncoupled simulations did not yield climatologically consistent sea-ice conditions or heat fluxes. When these flux corrections are continuously applied in a grid cell where the sea-ice conditions or the convection rate have changed during the coupled integration, unrealistic accumulations of ice or regions without any ice formation are the consequence.

The present sea-ice formulation with over-simplified dynamics and thermodynamics generally leads to sensitivities that are too high. The sea-ice model is therefore unable to 'buffer' the unrealistic forcing caused by model errors and the local inconsistencies arising from the flux corrections. The fact that the ice drift is determined by the currents of the uppermost layer of the OGCM leads to its severe underestimation in major parts of the divergent drift regions, specifically in the Antarctic ice pack, which is mostly wind driven (Stössel et al. 1990). As a result, the areal ice-thickness distribution (not shown) does not represent the observed dynamically-induced characteristics. Instead, the overestimated ice thicknesses are thermodynamically induced, being highly controlled by flux corrections of the order of 100 W/m^2.

The presently used sea-ice component of the coupled model has only a limited capability to simulate the present day climate. Predicted climate changes over regions with sea-ice coverage should therefore be treated with caution. We note, however, that the performance of the sea-ice model is comparable to that of other models used in climate change experiments (Houghton et al. 1992).

It cannot be excluded that the above-discussed systematic errors in the simulation of sea-ice lead to unrealistic responses in areas remote from the ice-covered regions. Since it is impossible to assess the impact of such errors, we assume that the first order effects are the same in the control and the climate change simulation.

Hydrological cycle. The simulation of the global mean hydrological cycle in the control experiment lies within the uncertainty range of the available observations (Baumgartner and Reichel 1975).

The evaporation and precipitation over land and sea is increased in the MC mean by about 1% during the 50 years (Fig. 12), which is more than a factor of two larger than the standard deviation between the MC experiments. However, it is still less than the uncertainty within the observations (Baumgartner and Reichel 1975). The increased evaporation over the oceans does not lead to a noticeably enhanced transport of water to land regions. About 25% more fresh waster is deposited on the permanent ice. This leads to a net fresh water loss of 1000 km^3/year, which would result in a lowering of global mean sea-level by 30 cm within 100 years. The control simulation already has a high water deficit of about 120 cm after 100 years, since the snow which falls onto ice-covered land areas is stored and cannot be recycled into the ocean (e.g., via calving of ice sheets, an effect which has not been parameterized). This deficit does not influence the ocean circulation, since the mean imbalance is automatically compensated by the flux correction (Cubasch et al. 1992).

The change in the precipitation pattern shows very little spatial coherence, regardless of whether definition 1 or definition 2 is used. This noisy appearance makes it difficult to separate a climate change signal from the model's internal variability (see Santer et al. 1993a). The standard deviation of the MC simulations (Fig. 13) is as large as the signal. A univariate *t*-test shows no spatially-coherent regions with precipitation changes significant at the 95% level. Our results suggest, therefore, that precipitation may not be a useful component of a multivariate "fingerprint" for early detection of greenhouse-gas-induced climate change. Nevertheless, given the importance of precipitation in terms of potential climate change impacts, we provide a brief description of the regional distribution of precipitation change.

The rainfall increases in the mid-latitudes of the Northern Hemisphere in northern winter (Fig. 13b) and in the monsoon region in northern summer (Fig. 13a), are consistent with the results from other climate models summarized in the IPCC report (Houghton et al. 1992). The increased rainfall along the east coast of the United States appears to be a stable feature in northern winter. In northern summer the North American continent experiences a decrease in precipitation, a feature which is strongly developed in only two of the four experiments. An increase in precipitation in the southern hemisphere storm-tracks, as suggested by IPCC (Houghton et al. 1992), cannot be identified as a stable feature.

The predicted climate change pattern for annual mean soil moisture is generally as noisy as the precipitation signal. In northern winter, however, there are soil moisture increases over wide areas of the Northern Hemisphere (Fig. 14b). This seems to be a stable phenomenon in all four MC integrations. India experiences an overall decrease in soil moisture.

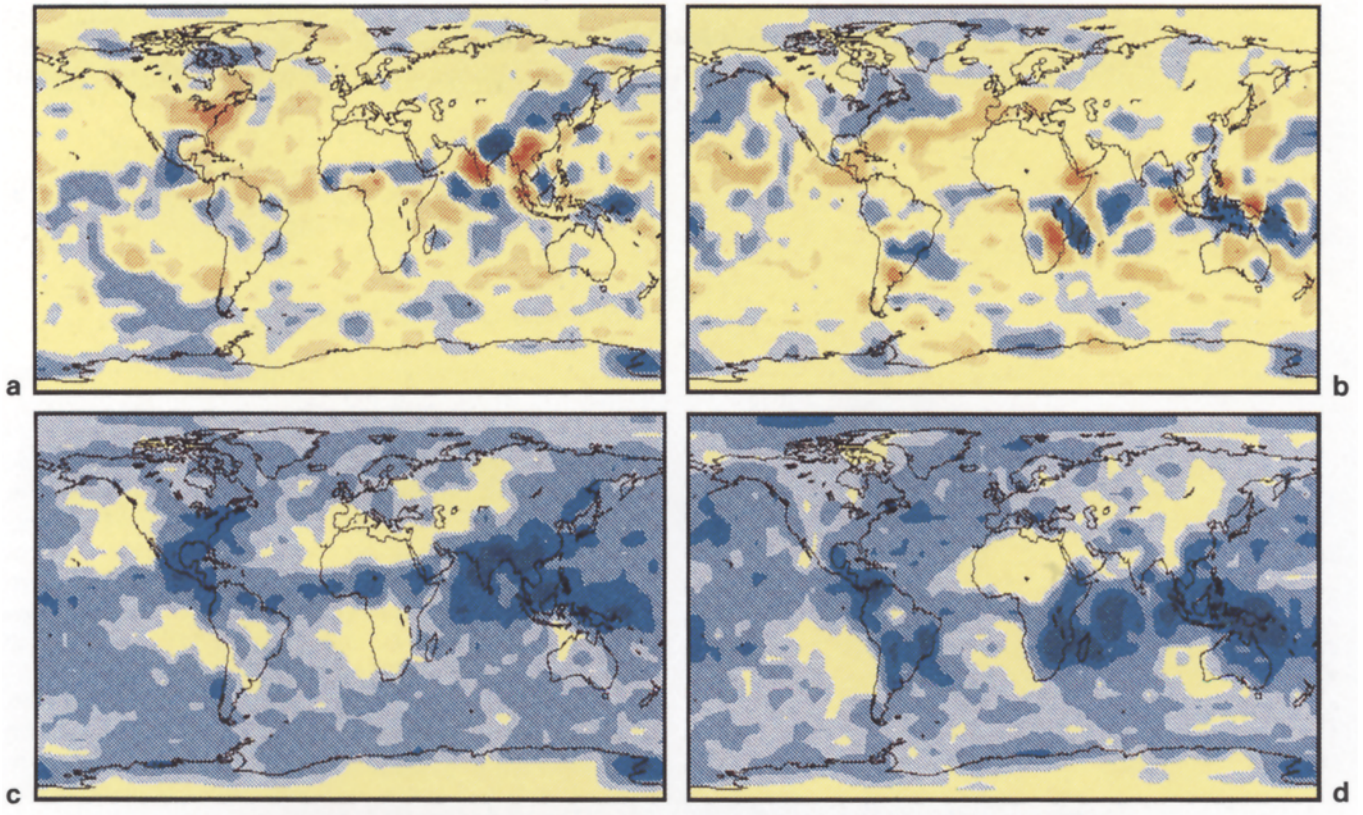
In the northern summer season the United States, India and Africa south of the equator generally become drier (Fig. 14a), in agreement with the IPCC (Houghton et al. 1992) results. These changes are not significant, as is evident from the large between-experiment standard deviations (Fig. 14c, d).

Sea level rise. The sea level changes presented in this section contain only the effects of thermal expansion and dynamic topography. The effect of glacier melting and the hydrological cycle are not included. The sea level rise has been calculated as the difference relative to the control run at the same year (definition 2), as in Cubasch et al. (1992), since the global mean sea level in the control integration shows a drift of about 1.5 cm during the first 50 years. Contrary to the behavior of the global mean temperature, the sea level rise after 50 years of integration time is almost identical in all four simulations (Fig. 15). This is due to the fact that the thermal inertia and integration effect of the ocean smooth out the short-term fluctuations seen in surface temperature. The average sea level rise in the MC calculations is 4.2 cm within 50 years, with an RMS error of the order of 0.2 cm.

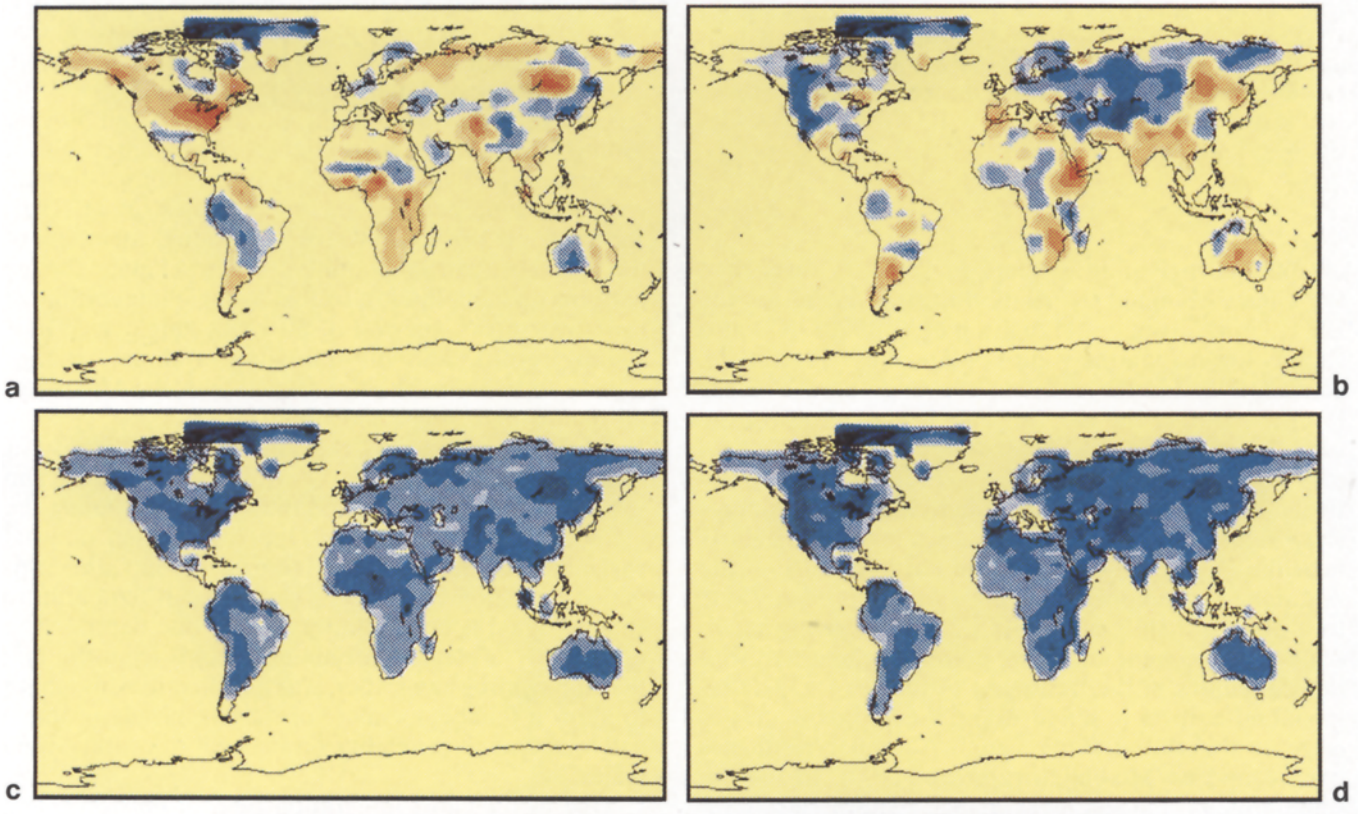
Following Hasselmann et al. (1993), a correction for the "cold start" has been calculated from the ensemble mean sea level rise. It is in the order of 3 cm within 50 years and represents a substantial additional contribution to the sea level rise (Fig. 15b). The rate of corrected sea level rise falls well within the bracket quoted in the IPCC report (Houghton et al. 1992). The

Fig. 13a–d. The regional distribution of changes in the ensemble averaged precipitation (definition 1) in the final decade of the MC integrations for **a** summer and **b** winter seasons. Also shown are maps of the "between-experiment" standard deviation of precipitation change for **c** summer and **d** winter seasons. Units are mm/day

Fig. 14a–d. The regional distribution of changes in the ensemble averaged soil moisture (definition 1) in the final decade of the MC integrations for **a** summer and **b** winter seasons. Also shown are maps of the "between-experiment" standard deviation of soil moisture change for **c** summer and **d** winter seasons. Units are cm


a
b
c
d
−5.00 −2.00 −1.00 −0.50 −0.20 −0.10 0.10 0.20 0.50 1.00 2.00 5.00 10.00 15.00

13


a
b
c
d
−5.00 −2.00 −1.00 −0.50 −0.20 −0.10 0.10 0.20 0.50 1.00 2.00 5.00

14

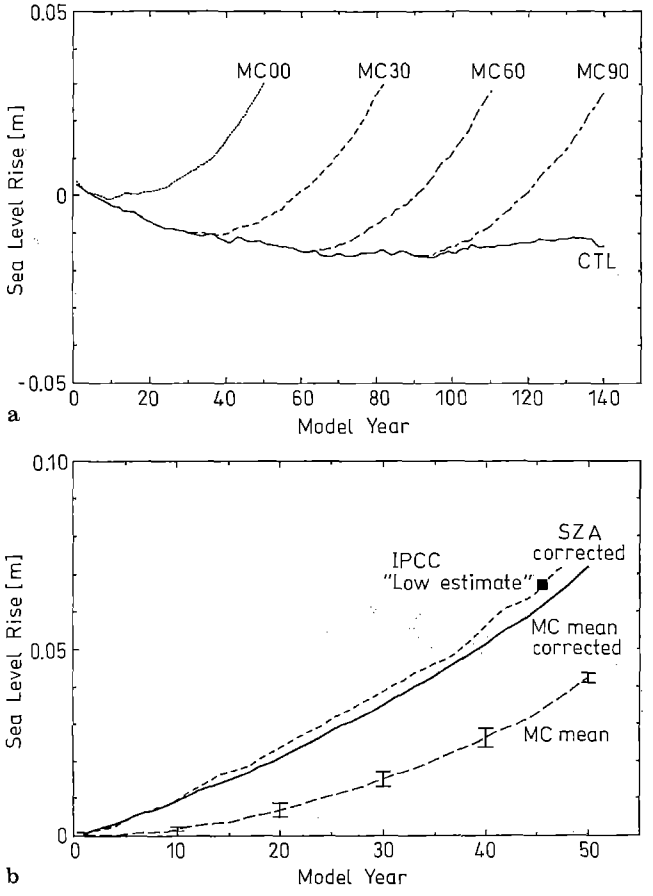


Fig. 15. **a** Time evolution of the global mean sea level change (cm) due to thermal expansion for the four Monte Carlo simulations and the control experiment. **b** Time evolution of the global mean sea level change (cm) for the ensemble mean, the cold start corrected ensemble mean, the cold start corrected scenario A calculation after Cubasch et al. (1992), the 95% confidence interval for the uncorrected ensemble, and the IPCC "low" estimate for the year 2035 after Houghton et al. 1990

absolute change after 50 years, however, is lower than the values obtained by a box diffusion model for the IPCC "low" estimate (Houghton et al. 1990). It is also slightly lower than the values obtained by Hasselmann et al. (1993) for the scenario A integration. The sea level rise during the first 50 years is about 25% of the value after 100 years.

The horizontal pattern of sea level rise is comparatively stable (Fig. 16). The individual experiments show similar regions of increased sea level, particularly near the Bering Strait and at the northern edge at the Antarctic Circumpolar Current (ACC), probably due to a poleward shift of the ACC. The relatively small-scale extreme values near the poles are connected with the difficulties in the treatment of the sea-ice. The regional differences in sea-level rise are approximately as large as the global mean signal. A similar result has been found by Mikolajewicz et al. (1990). In the ensemble mean, the sea level change pattern closely resembles the pattern obtained after 100 years of integration time, but with much reduced amplitude.

The sea level change exhibits a large between-integration variability only in the vicinity of the ACC and in regions with sea-ice. A univariate *t*-test (not shown) indicated areas of significant (95% level) sea-level change in the Indian Ocean, the north and central Pacific, and the northern edge of the ACC. In the Atlantic and the central and southern ACC only limited regions showed significant changes.

4 Summary and conclusions

Our study had the goal of investigating the statistical properties of the climate system's response to greenhouse-gas forcing. We therefore performed four time-dependent greenhouse warming experiments with the same global coupled atmosphere-ocean model, but with each simulation using initial conditions from different "snapshots" of the control run climate. The radiative forcing – the increase in equivalent CO_2 concentrations from 1985 to 2035 specified in the Intergovernmental Panel on Climate Change (IPCC) scenario A – was identical in all four integrations.

Despite the limitation of a very small sample size, this approach enables the estimation of both a mean response and the "between-experiment" variability, information which is not available from a single integration. The use of multiple realizations provides insights into the stability of the response, both spatially, seasonally, and in terms of different climate variables. The "between experiment" variability which we consider here may be due to inherent intransitivity of the climate system, internally-generated fluctuations of the coupled ocean-atmosphere system, residual climate drift, or some combinations of these factors.

Generally, the mean of the ensemble of four 50-year Monte Carlo (MC) experiments confirms the results of the original 100-year scenario A integration performed with the same coupled model (Cubasch et al. 1992; Santer et al. 1993a). However, individual members of the ensemble can show quite different climate change patterns and amplitudes, so that a single integration, at least over a time period of 50 years, might give misleading results. This illustrates that a climate change signal can be more clearly isolated with the MC method than from a single short integration.

In global mean terms, the climate change signals for near surface temperature, the hydrological cycle and sea level significantly exceed the variability among the members of the ensemble. Due to the high internal variability of the modelled climate system, the estimated detection time of the global mean temperature change signal is uncertain by at least one decade. The uncertainty in detection time also stems from the fact that the control integration was not of sufficient length to allow a reliable estimate of the magnitude of internally-generated variability on decadal-to-century time scales.

Due to the short integration time of only 50 years, there are both qualitative and quantitative differences in the temperature change patterns of the individual

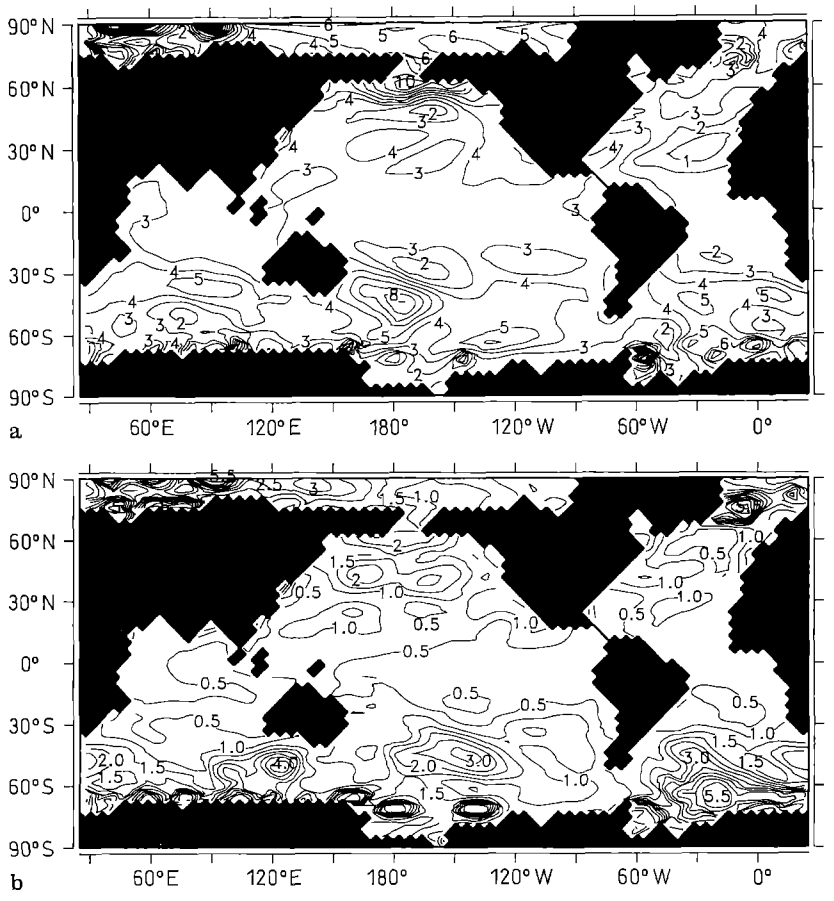


Fig. 16. a Horizontal distribution of the mean sea level change (cm) for the final decade of the ensemble mean and **b** the "between experiment" standard deviation of mean sea level change according to definition 2

members of the ensemble. This is consistent with the studies by Cubasch et al. (1992) and Santer et al. (1993a), where it was found that a single, dominant near surface temperature signal did not emerge until the second half of the 100-year scenario A integration. In the first half of this experiment, the surface temperature signal was relatively noisy and was composed of a large number of individual patterns.

While the ensemble mean surface temperature and sea level fields show significant responses to greenhouse-gas forcing (both locally and in overall terms), it is not possible to identify a significant response in the precipitation and soil moisture fields, variables which are spatially noisy and characterized by large variability between the individual integrations. This agrees with results obtained by Santer et al. (1993a), where changes in precipitation were difficult to separate from the coupled model's inherent climate variability and/or drift.

The seasonal patterns of the ensemble mean temperature and precipitation change agree reasonably well with the results published in the recent IPCC report (Houghton et al. 1992). An increase in precipitation in the storm tracks of the Southern Hemisphere, as suggested in the same IPCC report, could not be verified.

The individual realizations show little difference in the predicted global mean sea-level rise due to thermal expansion. The simulated changes are slightly lower than the lowest IPCC estimate, even when corrected for the "cold start" error. The cold start contributes substantially to the sea level rise in the time period considered. After 50 years integration time more than a third of the contribution to the sea level rise is estimated to arise from the cold start error.

The near-surface temperature signals in the 50-year MC experiments show a greater sensitivity to the precise signal definition than the temperature signals in the longer Cubasch et al. (1992) scenario A integration. The results can vary considerably, depending on the assumptions made regarding the cause of changes in the state of the control experiment – i.e. whether these are primarily considered as internal fluctuations or residual climate drift. In our study, the same experiment can yield either large-scale Arctic cooling or large-scale Arctic warming, simply by defining the signal in two different ways. A larger ensemble should reduce uncertainties and clarify whether an unusual response has been obtained by chance, or represents a more frequently emerging state. Furthermore, in order to make more reliable statements about high-latitude climate change from globally coupled ocean-atmosphere models, it will be necessary to incorporate a more sophisticated formulation of the sea-ice component and to reduce the size of the flux correction.

The MC simulations complement the longer 100-year scenario A experiment. While the long experiment was necessary to establish a stable climate change

pattern for near surface temperature, the MC simulations give an estimate of both the mean climate change pattern and its uncertainties on a time scale of 50 years.

This study has focussed on the uncertainties in a time-evolving climate change signal caused by our imperfect knowledge of the initial climate state. We note, however, that there are a variety of other sources of uncertainty which we do not consider here. These include sensitivity of experimental results to model resolution and to different parameterization schemes. For example, recent studies have shown that the implementation of different schemes for parameterizing cloud formation processes can lead to a wide range of climate sensitivities in response to greenhouse gas forcing (Mitchell and Ingram 1992). Additional sources of uncertainty are introduced by systematic model errors (e.g., those involving the simulation of sea-ice), by the "cold start" phenomenon, by the short integration time, and by unrealistically large flux corrections and incomplete flux compensation, which may lead to residual drift of the coupled system. The assessment of all factors influencing the uncertainties of our climate prediction is beyond the scope of this paper. We have shown however, that future greenhouse gas studies must address *both* the changes in mean state and the uncertainties associated with these changes.

Acknowledgements. We wish to thank K. Hasselmann, L. Bengtsson, E. Roeckner, H. Grassl and G. Boer for their helpful discussions and their support with scientific aspects of the paper. P. Lenzen and R. Gaedtke provided for the data handling. N. Noreiks and M. Grunert drew the diagrams. The research has been supported by the German Ministry for Research and Technology (BMFT), the Max-Planck-Gesellschaft, the Freie und Hansestadt Hamburg, the Deutsche Forschungsgemeinschaft (SFB 318) and the EC Environmental Program. The authors are grateful to the staff of the DKRZ for their excellent technical support.

References

Bakan S, Chlond A, Cubasch U, Feichter J, Graf HF, Grassl H, Hasselmann K, Kirchner I, Latif M, Roeckner E, Sausen R, Schlese U, Schriever D, Schult I, Schumann U, Sielmann F, Welke W (1991) Climate response to smoke from the burning oil wells in Kuwait. Nature 351:367–371

Baumgartner A, Reichel E (1975) The world water balance: mean annual global continental and maritime precipitation, evaporation and runoff. R Oldenbourg, München–Wien

Brankovic C, Palmer TN, Molteni F, Tibaldi S, Cubasch U (1990) Extended range predictions with the ECMWF models: time lagged ensemble forecasting. Q J R Meteorol Soc 116:835–866

Cubasch U, Hasselmann K, Höck H, Maier-Reimer E, Mikolajewicz U, Santer BD, Sausen R (1992) Time-dependent greenhouse warming computations with a coupled ocean-atmosphere model. Clim Dyn 8:55–69

Fichefet T, Tricot C (1992) Influence of the starting date of model integration on projections of greenhouse-gas-induced climatic change. Geophys Res Lett 19:1771–1774

Gloersen P, Campbell WJ (1988) Variations in the Arctic, Antarctic, and global sea ice covers during 1978 to 1987 as observed with the Nimbus 7 scanning multichannel microwave radiometer. J Geophys Res 93:10666–10674

Hansen J, Lacis A, Ruedy R (1990) Comparison of solar and other influences on long-term climate. In: Climate impact of solar variability, Conf. Publn 3086, NASA, Greenbelt, pp 149–159

Hasselmann K, Sausen R, Maier-Reimer E, Voss R (1993) On the cold start problem in transient simulations with coupled ocean-atmosphere models. Clim Dyn 9:53–61

Houghton JT, Jenkins GJ, Ephraums JJ (eds) (1990) Climate change. The IPCC scientific assessment. Cambridge University Press, Cambridge

Houghton JT, Callander BA, Varney SK (eds) (1992) Climate Change 1992. The supplementary report to the IPCC scientific assessment. Cambridge University Press, Cambridge

Livezey RE, Chen WY (1983) Statistical field significance and its determination by Monte Carlo techniques. Mon Weather Rev 111:46–59

Loewe P (1987) Sea ice simulations performed with forcing fields specified from a GCM. Techn. Rep. 6/87, Cold Regions Research Lab. 72, Hanover, New Hampshire

Lorenz EN (1975) Climatic predictability. In: The physical basis of climate and climate modelling. GARP publication series No. 16, WMO Geneva, pp 132–137

Lorenz EN (1984) Irregularity: a fundamental property of the atmosphere. Tellus 36A:98–110

Lorenz EN (1991) Chaos, spontaneous climatic variations and detection of the greenhouse effect. In: Schlesinger M (ed) Greenhouse-gas-induced climatic change: a critical appraisal of simulations and observations. Elsevier, Amsterdam, pp 445–453

Maier-Reimer E, Mikolajewicz U, Hasselmann K (1993) Mean circulation of the Hamburg LSG OGCM and its sensitivity of the thermohaline surface forcing. J Phys Oceanogr 23:731–757

Manabe S, Spelman MJ, Stouffer RJ (1992) Transient responses of a coupled ocean-atmosphere model for gradual changes of atmospheric CO_2. Part II: seasonal response. J Clim 5:105–126

Mikolajewicz U, Maier-Reimer E (1990) Internal secular variability in an ocean circulation model. Clim Dyn 4:145–156

Mikolajewicz U, Santer BD, Maier-Reimer E (1990) Ocean response to greenhouse warming. Nature 354:589–593

Mitchell JFB, Ingram WJ (1992) On CO_2 and climate: mechanisms of changes in cloud. J Clim 5:5–21

Molteni F, Cubasch U, Tibaldi S (1988) 30- and 60-day forecast experiments with the ECMWF spectral models. In: Chargas C, Puppi G (eds) Persistent meteo-oceanographic anomalies and teleconnections. Pontif Acad Sci Scripta Varia, MCMLXXXVIII, 505–555

Murphy JM (1992) A prediction of the transient response of climate. Climate Research Technical Note 32, Hadley Centre, Bracknell, UK

Parkinson CA (1991) Interannual variability of the spatial distribution of sea ice in the North Pole region. J Geophys Res 96:4791–4801

Parkinson CA (1992) Interannual variability of monthly southern ocean sea ice distributions. J Geophys Res 97:5349–5363

Parkinson CA, Bindschadler RA (1984) Response of Antarctic sea ice to uniform atmospheric temperature increases. Geophys Monogr Am Geophys Union 29:254–264

Santer BD, Brüggemann W, Cubasch U, Hasselmann K, Höck H, Maier-Reimer E, Mikolajewicz U (1993a) Signal-to-noise analysis of time-dependent greenhouse warming experiments. Part 1. Pattern analysis. Clim Dyn 9:267–285

Santer BD, Cubasch U, Mikolajewicz U, Hegerl G (1993b) The use of general circulation models in detecting climate change induced by greenhouse gases. PCMDI Report No. 10, Program for Climate Model Diagnosis and Intercomparison, Lawrence Livermore National Laboratory, Livermore

Sausen R, Barthel K, Hasselmann K (1988) Coupled ocean-atmosphere models with flux correction. Clim Dyn 2:154–163

Storch H (1982) A remark on Chervin-Schneider's algorithm to test significance of climate experiments with GCMs. J Atmos Sci 39:187–189

Stouffer RJ, Manabe S, Bryan K (1989) Interhemispheric asymmetry in climate response to a gradual increase of atmospheric CO_2. Nature 342:660–662

Stössel A, Lemke P, Owens WB (1990) Coupled sea ice-mixed layer simulations for the Southern Ocean. J Geophys Res 95:9539–9555

Washington WM, Meehl GA (1989) Climate sensitivity due to increased CO_2: experiments with a coupled atmosphere and ocean general circulation model. Clim Dyn 4:1–38

Wigley TML, Raper SCB (1990) Natural variability of the climate system and detection of the greenhouse effect. Nature 344:324–327

Wigley TML, Santer BD (1990) Statistical comparison of spatial fields in model validation, perturbation and predictability experiments. J Geophys Res 95:851–865

Zebiak SE, Cane MA (1987) A Model El Niño-Southern Oscillation, Mon Weather Rev 115:2262–2278

Zwally HJ, Comiso JC, Parkinson CL, Carsey FD, Campbell WJ, Gloersen P (1983) Antarctic sea ice 1973–1976: satellite passive-microwave observations, NASA Spec Publ SP-459:1–206